

Alberto J. Lopez

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EXPERIENCE

Quality Engineer at Rennscot MFG

January 2025 - Present

- Designed process, programmed, and operated multiple Zeiss CMMs (Contura, Micura, and Duramax) for the inspection of highly complex aerospace components under AS9100 standards.
- Conducted process capability studies to reduce inspection frequencies for multiple components.
- Maintained and organized the uploading/management of calibration paperwork for most inspection equipment.
- Performed and oversaw incoming inspection of hard-anodized and chem-filmed aerospace electro-mechanical components.
- Handled customer communications regarding quality control, inspection planning, and helped quote jobs.
- Provided feedback for machining changes based on geometric inspection data to machinists.

Software Engineer at Pratt & Whitney

June 2023 - December 2024

- Designed and implemented flexible, scalable data analysis tools for neural net models, contributing to improved process efficiency.
- Designed and implemented auditing tools for our neural net models, and their deployment environments.
- Collaborated with cross-functional teams using Agile methodologies to drive product development, ensuring alignment with project timelines and goals.

Process Development Engineer at Artifex Solutions LLC

May 2022 - June 2023

- Designed and implemented automated quality control processes for Class II medical devices, including orthopedic implants.
- Developed custom fixturing for CMM, laser marking, and vision quality tools using GD&T principles, managing the entire process from design to manufacturing as design engineer.
- Led the development of a CMM quality process using a Zeiss Contura, reducing QC turnaround time from 10 days to 1 hours per batch.
- Applied GD&T standard ASME Y14.5 for detailed manufacturing drawings and fixture design.

EDUCATION

M.S. in Engineering Management - Currently Enrolled

Northeastern University - Started 2025

B.S. in Mechanical Engineering - Aerodynamics Concentration

The Catholic University of America — August 2018 - May 2022

SKILLS

CAD: SolidWorks,, Onshape, Fusion360

Manufacturing Processes: CNC machining, 3D printing, CMM programming

Metrology: Calipers, Micrometers, Height Gauge, Zeiss Calypso Certified (Advanced Training)

Design: GD&T, fixture design, mechanical systems design

Analysis: pFMEA, Gage R&R, Test Method Validation

Programming: Python, Javascript, SQL, Java, bash

INTERESTS

Running: Iceland Marathon 2024, Double Blossom Race DC 2023/24/25, Cambridge/Boston Half Marathon 2024/25

Watersports: Wing-surfing, sailing, scuba diving (PADI Open Water Diver)